

Homework 06: Chapter 07 part 2

EEC 3150

For each Python exercise, write a separate Python script. Submit all scripts on Blackboard under the appropriate assignment. Name your script files with the following format:

HW<assignment number>_Ex<exercise number>_<FirstInitial><LastName>.py

Example: HW01_Ex01_GWest.py

For short answer or math exercises, either do your work on a sheet of paper and upload a picture of it to Blackboard or submit some kind of text file or Word document to Blackboard. You can place all such exercises in a single document as long as you clearly label each exercise. These files should follow a similar naming convention as the Python scripts.

Exercise 1 (10 points, Python)

This exercise will require you to write two `.py` files.

1. Write a module called `MyModule.py` that defines several functions and variables of your choice.
2. Write a program which will import and call each of these functions and variables. This file should have our standard homework naming convention.

Exercise 2 (15 points, Python)

1. Write a program which will obtain a list of files in the `cwd` using `os.listdir()` and create a dictionary whose keys are filetypes and whose values are a list of files of that type (all files are taken from the list of files return by `os.listdir()`). For example, if the files `a.txt`, `b.txt`, `c.py` were the in directory, the program should create the following dictionary:

```
{ 'txt': ['a.txt', 'b.txt'],  
  'py': ['c.py'] }
```
2. Print the keys and values neatly.
3. Write the dictionary to a file called `FileTypes.pk` using `pickle`.

Exercise 3 (20 points, Python)

1. Write two functions that perform the same task in different ways. The task should take anywhere between (roughly) 0.1 and 1 seconds to complete (basically it needs to take a while).
2. Use the `time` module to find the average time it takes to perform each function. Clearly print average times, clearly label each version.

Exercise 4 (30 points, Python)

In this exercise, you will calculate the ages of several actors using the `re`, `time` modules. See the `Birthdays.txt` file. The dates are formatted like `mm/dd/yyyy`.

1. Write two functions that will each return a dictionary whose keys are the names of actors and whose values are their birthdays. The first should use regular expressions to extract the names and dates (name it `WithRegex(fname)`) and the other should NOT use regex (call it `WithoutRegex(fname)`).
2. Write a function `GetAge(today, bday)` which will take today's date (formatted in the same manner as above) and an actor's birthday and return their age as an integer number of years. NOTE: you should take into account whether or not their birthday has occurred yet as of the `today` date and not merely subtract the years.
3. Use these functions to write the ages of the actors to a file `Ages.txt` with the following format (try to use formatted strings):
`Jane Doe is 10 years old`